
This content is from the eCFR and is authoritative but unofficial.

Title 33 — Navigation and Navigable Waters

Chapter II — Corps of Engineers, Department of the Army, Department of Defense

Part 234 Corps of Engineers Agency Specific Procedures to Implement the Principles, Requirements and Guidelines for Federal Investments in Water Resources

- § 234.1 General.
- § 234.2 Definitions.
- § 234.3 Exceptions.
- § 234.4 Objectives and applicability.
- § 234.5 Level of analysis.
- § 234.6 The planning process.
- § 234.7 Evaluation framework.
- § 234.8 Final array of alternatives.
- § 234.9 Evaluate effects of alternatives.
- § 234.10 Compare alternatives.
- § 234.11 Select the recommended plan.

PART 234—CORPS OF ENGINEERS AGENCY SPECIFIC PROCEDURES TO IMPLEMENT THE PRINCIPLES, REQUIREMENTS AND GUIDELINES FOR FEDERAL INVESTMENTS IN WATER RESOURCES

Authority: 42 U.S.C. 1962-3.

Source: 89 FR 104021, Dec. 19, 2024, unless otherwise noted.

§ 234.1 General.

- (a) This part prescribes the Agency Specific Procedures (ASPs) for the United States Army Corps of Engineers (Corps) to execute its Civil Works mission, in accordance with the Water Resources Principles and Guidelines defined in section 2031 of the Water Resources and Development Act (WRDA) of 2007 (Pub. L. 110-114; 42 U.S.C. 1962-3), the Principles, Requirements and Guidelines (PR&G) issued by the Council on Environmental Quality and approved by the Water Resources Council, and as called for in section 110 of WRDA 2020 (Division AA of Pub. L. 116-260).
- (b) Section 2031 of the WRDA of 2007 (Pub. L. 110-114) directed the Secretary of the Army to revise the March 10, 1983, Economic and Environmental Principles and Guidelines for Water and Related Land Resources Implementation Studies (P&G) for Corps use and to address the following considerations:

advancements in economic and analytic techniques; public safety; low-income communities; nonstructural approaches; interaction with other water resources projects and programs; integrated and adaptive management; and use of public benefits to justify projects. This WRDA provision also provided that the Federal Objective is to reflect national priorities, encourage economic development, and protect the environment by seeking to maximize sustainable economic development, avoid the unwise use of floodplains, and protect and restore natural ecosystems.

- (c) The PR&G was issued as an interagency effort to modernize the P&G. The PR&G is comprised of the Principles and Requirements (P&R) issued in March 2013 and the Interagency Guidelines issued in December 2014. The PR&G emphasizes that water resources projects should strive to meet the Federal Objective and maximize public benefits relative to public costs. The PR&G is designed to support water infrastructure projects with the greatest public benefits (economic, environmental, and social benefits) relative to costs.
- (d) Congress directed the Secretary of the Army to issue ASPs to implement the PR&G in section 110 of WRDA 2020 (Division AA of Pub. L. 116-260).

§ 234.2 Definitions.

Acceptability. The viability and appropriateness of an alternative from the perspective of the Nation's general public and consistency with existing Federal laws, authorities, and public policies. It does not include local or regional preferences for solutions or political expediency.

Adaptive management. A deliberate, iterative, and scientific-based process of designing, implementing, monitoring, and adjusting an action, measure, or project to address changing circumstances and outcomes, reduce uncertainty, and maximize one or more goals over time.

Completeness. The extent to which an alternative provides and accounts for all features, investments, and/or other actions necessary to realize the planned effects, including any necessary actions by others. It does not necessarily mean that alternative actions need to be large in scope or scale.

Effectiveness. The extent to which an alternative alleviates the specified problems and achieves the specified opportunities.

Efficiency. The extent to which an alternative alleviates the specified problems and realizes the specified opportunities at the least cost.

Federal investment. Investments made by the Corps related to water resources development projects, including flood and storm risk management, ecosystem restoration, land management activities, navigation, recreation, and hydropower.

Federal Objective. The fundamental goal of Federal investments in water resources. Federal water resources investments shall reflect national priorities, encourage economic development, and protect the environment. Federal investments should strive to maximize net public benefits.

Indigenous Knowledge. Indigenous Knowledge may be described as a body of observations, oral and written knowledge, innovations, practices, and beliefs developed by Tribes and Indigenous Peoples through interaction and experience with the environment, consistent with the definitions used in 43 CFR 2361.5 and 6101.4(h) and the Guidance for Federal Departments and Agencies on Indigenous Knowledge, issued November 30, 2022. It is applied to phenomena across biological, physical, social, cultural, and spiritual systems. Indigenous Knowledge can be developed over millennia, continues to develop, and includes understanding based on evidence acquired through direct contact with the environment and long-term experiences, as well as extensive observations, lessons, and skills passed from generation to generation.

Nature-based alternatives. An alternative comprised of actions to protect, sustainably manage, or restore natural or modified ecosystems to address societal challenges, while simultaneously providing benefits for people and the environment.

Non-Federal interest.

- (1) A legally constituted public body (including an Indian Tribe and a Tribal Organization (as those terms are defined in 25 U.S.C. 5304)); or
- (2) A nonprofit entity with the consent of the affected local government, that has full authority and capability to perform the terms of its agreement and to pay damages, if necessary, in the event of failure to perform.

Nonstructural alternative. An alternative comprised of a nonstructural approach or combination of nonstructural approaches that addresses the water resources problem.

Nonstructural approach. An approach that alters the use of existing infrastructure or human activities to generally avoid or minimize adverse changes to existing hydrologic, geomorphic, and ecological processes. This may include measures such as certain forms of nature-based solutions; modified floodplain practices; policy modifications; vessel speed limits; traffic management and tidal navigation restrictions; the reoperation of dams and reservoirs to restore or better mimic natural hydrology and flow patterns; invasive plant removal; signage to limit public access at an aquatic ecosystem restoration site; setbacks; elevations; relocation; buyout/acquisition including the acquisition of flowage easements; dry flood proofing; and wet flood proofing. They may also include actions that are not the responsibility of the Corps such as providing flood insurance, establishing building codes for new construction, and other local floodplain management practices, installing early warning systems, and developing emergency evacuation plans.

Professional judgment. An evidence-based decision that relies on appropriate training and experience.

Public benefits. Encompasses economic, environmental, and social impacts, and includes those that can be quantified in monetary terms, as well as those that can be quantified or described qualitatively.

Regional economic development effects. The changes in the distribution of regional economic activity that would result from implementation of an alternative plan. These economic effects amount to a transfer of resources from one part of the Nation to another (either from one region of the country to another, or within a region). They accrue in a local area or region but are offset by equivalent losses elsewhere in the country.

Regulatory. Generally, those activities subject to legal restrictions promulgated by the Federal Government.

Resilience. Resilience is the ability to prepare for threats and hazards, adapt to changing conditions, and withstand and recover rapidly from adverse conditions and disruptions.

Sustainable. The creation and maintenance of conditions under which humans and nature can coexist in the present and into the future.

Tribal Nation (Federally recognized Indian Tribe or Tribal Organization). An Indian or Alaska Native tribe, band, nation, pueblo, village, or community that the Secretary of the Interior acknowledges to exist as an Indian Tribe pursuant to the Federally Recognized Indian Tribe List Act of 1994, 25 U.S.C. 5130.

Unwise use of floodplains. Any action or change that diminishes public health and safety, or an action that is incompatible with or adversely impacts one or more floodplain functions that leads to a floodplain that is no longer self-sustaining or degrades ecosystem services.

Watershed. A land area that drains to a common waterbody.

§ 234.3 Exceptions.

Exceptions to any requirements or policy contained in this part may be requested by the Corps or the non-Federal interest or responsible Tribal, State, or local government. Exceptions must be requested in writing and will be reviewed for a decision by the Assistant Secretary of the Army for Civil Works.

§ 234.4 Objectives and applicability.

- (a) **Introduction.** The goal of the Department of the Army's ASPs is to ensure that Army Civil Works consistently applies a common framework for analyzing a diverse range of water resources development projects, programs, activities, and related actions involving Federal investments. The ASPs will advance transparency and consistency of the Corps' Federal investments in water resources. The intention of the ASPs is to outline the steps to apply the PR&G to Corps water resources investments, including a determination of the applicability of the PR&G in the context of the Corps' missions and authorities, to provide a common framework for evaluation of investment alternatives, and to ensure that the Corps adequately addresses the Guiding Principles identified in the P&R.
- (b) **Objectives for Federal water resources investments.** Section 2031 of WRDA 2007 (Pub. L. 110-114; 42 U.S.C. 1962-3) specifies that Federal water resources investments shall reflect national priorities, encourage economic development, and protect the environment. The Corps shall accomplish this Federal Objective of water resources planning policy by:
 - (1) Seeking to maximize sustainable economic development;
 - (2) Seeking to avoid the unwise use of floodplains and flood-prone areas and minimizing adverse impacts and vulnerabilities in any case in which a floodplain or flood-prone area must be used; and
 - (3) Protecting and restoring the functions of natural systems and mitigating any unavoidable damage to natural systems.
- (c) **Net public benefits.** The Corps shall strive to maximize net public benefits to society. Public benefits encompass economic, environmental, and social goals, include monetized and un-monetized effects, and allow for the consideration of both quantified and unquantified effects.
- (d) **Applicability.**
 - (1) The objectives in paragraph (b) of this section shall be embodied in all new Army Civil Works' water resources investments, which include both structural and nonstructural approaches to water resources problems. The PR&G analysis under the Corps' ASPs described in this part is generally required for feasibility studies; general re-evaluation reports; major rehabilitation reports; studies performed under the continuing authorities program of the Corps; studies to support significant changes to project operations including any such changes that warrant preparation of an environmental impact statement, re-allocation studies, and studies conducted under section 216 of the Flood Control Act of 1970 (33 U.S.C. 549a); and any other project or program not otherwise excluded under paragraph (d)(2) of this section.
 - (2) The PR&G is not intended to apply to all Federal actions. The following types of Federal investments are identified as excluded from the requirements of this part:
 - (i) Regulatory actions, such as the issuance of permits associated with section 404 of the Clean Water Act (33 U.S.C. 1344).

- (ii) Real estate actions.
- (iii) Planning Assistance to States program.
- (iv) Flood Plain Management Services program.
- (v) Section 14 of the Rivers and Harbors Act of 1899 (33 U.S.C. 408) program.
- (vi) Public Law 84-99 program.
- (vii) Water Infrastructure Finance and Innovation Act Program.
- (viii) Environmental Infrastructure projects.
- (ix) Land management plans.
- (x) Operation and maintenance activities that are carried out in a manner consistent with the existing approved operation and maintenance manual or plan for an authorized project. This exclusion does not apply to significant proposed changes to an existing O&M plan including proposals that may be controversial, significant changes to the existing plan to meet new goals, and other significant changes that may warrant a further analysis of the options for operation and maintenance.
- (xi) International and Interagency Services and Support for Others actions.
- (xii) Research or monitoring activities.
- (xiii) Emergency actions.
- (xiv) Projects, programs, or plans that fall below the thresholds identified in table 1 to § 234.5(c). These excluded actions generally occur when investments are routine and have inconsequential effects on water resources.
- (xv) Additional programs, plans, or projects which the Assistant Secretary of the Army for Civil Works determines do not require analysis pursuant to § 234.3.

§ 234.5 Level of analysis.

- (a) ***Standard and scaled levels of analysis.*** Once a determination has been made that the PR&G does apply, the level of analysis shall be determined. The level of PR&G analysis required will vary in scope and magnitude across programs and activities. There are two levels of analysis: standard and scaled. In general, the level of analysis should be commensurate with the significance of the Federal investment in terms of dollar value and the potential environmental impacts. While there is not a clear distinction between the different levels of analysis, the two types of analysis can generally be distinguished in several ways:
 - (1) A standard analysis seeks to evaluate all the relevant benefits and costs associated with the project or activity using original or secondary data. This type of analysis is typically used for new or significantly modified actions. The Corps would conduct a benefit-cost analysis of programs and activities that have some effect on the environment. For projects/activities that fall into the category of standard analysis, the analysis should make significantly greater efforts to quantify and monetize impacts. The extent to which effects can and should be monetized should be made on a resource-by-resource basis and should consider the estimated present value cost of the project/activity and the significance of the effects.

- (2) A scaled analysis is an analysis that is more limited in scope for projects, programs, or plans that have low risk/low cost, have minimal consequences of failure, pose minimal threats to human life or safety, or do not result in significant impacts to the environment. A scaled analysis may rely on benefits function transfer methods and readily available secondary data sources. Benefits function transfer methods are used to estimate monetary values by transferring available information about relationships from studies already completed to another location, context, or issue. Best practices would be applied when using this approach to avoid common pitfalls.
- (b) **Determining the appropriate level of analysis.** In many cases, professional judgment and available resources will be important factors in determining the appropriate level of analysis. The Corps will ensure that cumulative effects of many small, routine actions would not in itself elevate those investments to a scaled or standard analysis. Many of those small, routine actions would be excluded from PR&G analysis.
- (c) **Scope and magnitude of analysis required.** The threshold criteria for project, programmatic, and individual plan level analysis for Army Civil Works is shown in table 1 to this paragraph (c). These thresholds represent guidelines for the level of analysis that is likely to be most appropriate for an activity, given the level of investment in, appropriations for, or cost of that activity. In determining whether a given activity or project falls under or exceeds the financial thresholds, it is the level of the present value of Federal investment that is the relevant criterion to use. However, for a particular activity, a different level of analysis may be more appropriate, and projects/programs may depart from these guidelines where such a departure is justified. In general, a scoping effort should be undertaken to evaluate the level of effort needed to analyze the full range of potential effects. Project-level analysis should generally be used for water resources investments when the Corps has discretion in site-specific investment decisions. A programmatic-level analysis generally has a broader scale and/or scope than a project-level analysis. Programmatic-level analysis generally relates to funding programs or where a proposal for a set of similar actions analyzed under one decision document may occur.

TABLE 1 TO PARAGRAPH (c)—MONETARY THRESHOLD CRITERIA

Type of activity	Total investment (\$M)	Annual federal appropriations (\$M)	Level of analysis
Projects	>30		Standard analysis.
Proposed Corps investments in water resources, such as infrastructure, ecosystem restoration, new construction, modifications or replacements to existing facilities, and operations and maintenance	15-30		Scaled analysis.
	<15		Excluded.
Programs		>100	Standard analysis.
		50-100	Scaled analysis.
		<50	Excluded.
Individual Plans		>50	Standard analysis.
Management plans, such as watershed, master, etc.		10-50	Scaled
		<10	

Type of activity	Total investment (\$M)	Annual federal appropriations (\$M)	Level of analysis
			analysis. Excluded.

§ 234.6 The planning process.

- (a) **Introduction.** The following planning process will be used to implement the common framework summarized in the Interagency Guidelines for analyzing Federal investments in applicable water resources. The planning process will ensure that plan formulation, evaluation, and recommendations for proposed Corps investments reflect the Guiding Principles identified in the P&R: healthy and resilient ecosystems, sustainable economic development, floodplains, public safety, environmental justice, and a watershed approach. The planning process consists of a series of steps that identify or respond to problems and opportunities, as well as specific Tribal, State, and local concerns, and, in most cases, culminates in a recommended plan. The process involves an orderly and systematic approach to making determinations and decisions at each step so that the interested public and decision-makers in the planning organization can be fully aware of the following: the basic assumptions employed; the data and information analyzed; the areas of risk and uncertainty; the reasons and rationales used; and the significant implications of each alternative. The Corps will identify impacts to Tribal treaty and water rights at the earliest phases and throughout the plan evaluation process, screening alternatives that impact Tribal treaty and water rights. The planning process is iterative to adapt to new information and understanding. The result of the planning process is investment advice. The advice may be a recommended plan or plans that seek to maximize net public benefits in addressing the identified water resources problem and a description of the analysis of the benefits and costs of that and other potential plans.
- (b) **National Environmental Policy Act (NEPA).** Where Federal investments in water resources require analysis under NEPA and this part, Army Civil Works should integrate, to the extent possible, the analysis in this part into existing planning processes, and may integrate this part and NEPA analyses in a single analytical document that reflects both processes. Army Civil Works shall seek opportunities to integrate other required Federal and State environmental reviews with their combined analyses.
- (c) **Guiding Principles.** The Guiding Principles provide the overarching concepts that the Corps seeks to promote through investments in water resources.
 - (1) **Environmental justice.** Environmental justice refers to the just treatment and meaningful involvement of all people regardless of income, race, color, national origin, Tribal affiliation, or disability, in agency decision-making and other Federal activities that affect human health and the environment so that people:
 - (i) Are fully protected from disproportionate and adverse human health and environmental effects (including risks) and hazards, including those related to climate change, the cumulative impacts of environmental and other burdens, and the legacy of racism or other structural or systemic barriers; and

- (ii) Have equitable access to a healthy, sustainable, and resilient environment in which to live, play, work, learn, grow, worship, and engage in cultural and subsistence practices. Environmental justice shall be considered throughout the Civil Works program and in all phases of project planning and decision-making. Army Civil Works projects and programs shall advance equity by meeting the needs of communities, such as by reducing disparate environmental burdens, protecting Tribal treaty rights, removing barriers to participation in decision-making, and increasing access to benefits provided by Civil Works programs, including for disadvantaged communities. The planning process shall put these communities at the front and center of studies, providing robust opportunities for effective participation in the planning and decision-making processes. Any disproportionate adverse public safety, human health, or environmental burdens of project alternatives on communities with environmental justice concerns shall be avoided, minimized, or mitigated to the greatest extent reasonable. The Corps shall ensure that communities with environmental justice concerns have meaningful opportunities to identify potential alternatives, effects, and mitigation measures. The Corps shall also be transparent in fully displaying, disclosing and considering the potential effects of alternative actions on communities with environmental justice concerns.
- (2) **Floodplains.** All future Federal investments in and affecting floodplains must meet some level of floodplain resilience. Alternatives affecting floodplains should aim to improve floodplain resilience if possible and also should avoid the unwise use of floodplains. To identify floodplain areas for the purpose of this section, the Corps will use the best-available and actionable science including a climate-informed science approach. If construction in the floodplain or adverse impacts to a floodplain's natural and beneficial functions cannot be avoided, then the alternative must minimize adverse impacts to these areas and mitigate unavoidable impacts using nature-based approaches where possible. The Corps shall identify and communicate potential adverse effects on floodplain functions for the various alternatives under consideration. Where the Corps proposes to construct a project feature in a floodplain because that is the best way to achieve flood risk reduction or other public purposes, that proposed Corps project is not automatically considered an unwise use of the floodplain. The Corps shall strive to sustain the floodplain's natural and beneficial functions to the maximum extent practicable given the project's purpose and need.
- (3) **Healthy and resilient ecosystems.** Alternatives shall protect the existing functions of ecosystems and may restore the health of damaged ecosystems to a less degraded and more natural state where feasible and in accordance with current study and cost-sharing authorities. When adverse environmental impacts cannot be completely avoided, alternatives shall strive to minimize environmental impacts. When a particular alternative will cause unavoidable damage to the environment, mitigation to offset damages shall be incorporated into that alternative and evaluated as part of that alternative. In developing alternatives, consideration shall be given to ecosystem resilience, including acknowledging the value of ecosystem services to people. When evaluating alternatives, the health of the affected ecosystem shall be measured in its current condition as the baseline and projected under the alternatives being considered, including the No Action alternative.
- (4) **Public safety.** Alternative solutions shall strive to avoid, reduce, or mitigate significant risks to public safety, including both loss of life and injury, and shall include measures to manage and communicate the residual risks. The impact and reliability of alternatives on significant risks to public safety must be evaluated for both existing and future conditions, considered in decision-making, and documented.

- (5) ***Sustainable economic development.*** The Corps' investments in water resources shall encourage sustainable economic development. This is accomplished through the sustainable use and management of water resources, ensuring overall water resources resilience. Sustainable economic development creates and maintains conditions under which humans and nature can coexist. Analysis under sustainable economic development shall present, where feasible, information about the environmental resources in the project area or the area where activities are occurring, and how the resources and their value might be expected to change over time. Physical capital information may also be included where relevant. Analysis shall also include information on socio-economic conditions under current and projected conditions. Economic, social, and environmental effects and benefits shall be incorporated into the analysis of alternatives.
- (6) ***Watershed approach.*** When developing alternatives, the water resources problem being addressed should be analyzed on a watershed-based level to facilitate inclusion of a complete range of solutions, after considering the breadth of impacts across the watershed. A key aspect of the watershed approach is the analysis of information regarding watershed conditions and needs, allowing for consideration of upstream and downstream conditions and needs; consideration of other projects and actions in place, underway or planned by other agencies within the watershed; and the more thorough addressing of the potential impacts of a proposed action. The scale of the watershed used to develop alternatives can vary. The appropriately sized watershed for the particular need being addressed shall be a case-specific determination based on the relevant facts and circumstances. The watershed scale used to develop alternatives should encompass a geographical area large enough to ensure plans address cause and effect relationships among affected resources and activities, both upstream and downstream, and cumulative in nature, that are important to gaining public benefits or avoiding harm from the project. The watershed approach ensures that the interconnectedness of systems is evaluated to fully understand the root causes and symptoms of the water resources problem and the full range of potential public benefits. Communication starting in the scoping phase with other agencies or Tribal, territorial, State, and local government partners working in the watershed will help realize a watershed approach. In addition, other potential investments in the watershed shall also be accounted for under the watershed approach.

(d) ***Collaboration.***

- (1) The planning process will seek to achieve full collaboration with a wide range of affected Tribes, governmental and non-governmental stakeholders, communities with environmental justice concerns, and the public in all stages of the planning process. Collaboration with Tribes, governmental and non-governmental stakeholders, communities with environmental justice concerns, and the general public throughout the planning process allows consideration of multiple perspectives and information sources, such as Indigenous Knowledge, and shall be emphasized throughout the planning process. Collaboration with Tribes, communities, and local and State governments is a critical element to help identify specific problems, opportunities, and significant constraints within the study area, and to help establish planning goals and objectives that are consistent with the objectives of this part and are locally appropriate. Starting at the earliest phase in the planning process, Tribes and other communities with environmental justice concerns shall have an opportunity to play a key role in identifying alternatives, enhancing the positive benefits to their communities from potential Federal investment, and describing any concerns they may have with a potential project. Such early, meaningful, and robust engagement will help identify and address problems, possible solutions, and scope studies. Robust, early collaboration with Tribes

does not negate the need for Tribal consultation. Robust, early collaboration with Tribes is in addition to the requirement to conduct early, meaningful, and robust Government-to-Government consultation with Tribal Nations when appropriate.

- (2) To improve Federal decision-making and to promote transparency, Army Civil Works shall seek to meaningfully collaborate with other Federal and non-Federal entities. Engagement methods and scope of engagement will depend on the stage of the planning process, the issues, and the groups that will be contributing ideas and information to the planning process, and shall use best practices and techniques for engagement. Engagement strategies shall consider Corps, Tribal, and community resource constraints. Indigenous Knowledge, information from Tribal Nations, local and State governments, non-governmental organizations, and the public shall be incorporated into the problem definition and forecasting of future conditions as well as the development and analysis of alternatives. Robust engagement and transparency throughout the planning process, including during the evaluation and comparison of alternatives, will help deliver sound investment advice for water resources solutions that maximize net public benefits.
- (e) **Investigations and data collection.** Investigations, data collection, and analysis should be ongoing and integrated early in the planning process. Investigations should be relevant to the planning objectives and constraints. The interdisciplinary study team should identify the most important areas to focus on in the study, such as: engineering and design; surface water and groundwater hydrology; hydraulics; geology; operations; water quality; land resources; power generation and conservation; economics; financing; environmental, social, and cultural impacts and mitigation; opportunities for recreation; cost estimation for construction, operation, maintenance, replacement, and energy consumption; and climate change (to include greenhouse gas emissions). Investigation, data collection, and analysis should leverage and incorporate information from Tribal, State, local, non-governmental sources, and the public. Additional investigations should be performed as necessary.
- (f) **Identify purpose, problems, needs, and opportunities.** To identify purpose, problems, needs, and opportunities, the Corps shall:
 - (1) Ensure that the planning goals and objectives are consistent with the study authority.
 - (2) Clearly identify the purpose of the study, the role of the Federal Government, as well as the views of the non-Federal interest (if any), cooperating agencies, Tribes, various stakeholders, and the public.
 - (3) Describe the problems and opportunities to which the agency is responding in a manner that will not foreclose consideration of the full range of reasonable alternatives, including options that the non-Federal sponsor may not support.
 - (4) Define the study area, including activities within the watershed that are relevant to the proposed project and areas where impacts should be avoided.
 - (5) Describe the plans for stakeholder involvement.
 - (6) Prepare a summary of the planning objectives and constraints to be used in the analysis of the Federal investment. This summary should include a discussion of stakeholder, partner, and public input.
 - (7) Include a discussion of the social and cultural context of the region and resources.

- (g) ***Inventory existing resources and forecast future conditions.*** A summary of the specific economic, environmental, and social setting within the study area shall cover the condition and functional relationships of affected resources; their development potential and possible conflicts in producing affected ecosystem services; and the local situation with respect to investment, climate, markets, affected communities, and basic economic productivity.
- (1) The phrase “forecast future conditions” generally relates to the identification of impacts associated with the alternatives, including the No Action alternative. Future conditions should be assessed and analyzed as part of the evaluation process, and the best available data and forecast should be used to complete an analysis of these uncertain conditions.
 - (2) This exercise of identifying existing resources and forecasting future conditions will quantify, to the extent practicable, relevant water and related resource conditions as they currently exist within the study area, and forecast future conditions over the period of analysis. This would also include resources and conditions regarding the economic, environmental, and social aspects within the study area, as well as ecosystem services and climate-related scenarios. The existing resources and future conditions will be established using generally accepted sources that are national, State, or regional in scope, such as from peer-reviewed sources or sources which are government-produced.
 - (3) The “without-project condition” is the most likely condition expected to exist in the future over the period of analysis in the absence of a Federal investment by the Corps (through the proposed Corps project, action, or program that is under consideration), given current laws, policies, projects under construction, and any existing resources/conditions. It considers expected actions that may be executed by others, including potential future land use conditions, and shall consider effects of climate change using multiple scenario analyses.
 - (4) The “with-project condition” is the most likely condition expected to exist in the future, over the period of analysis, with a specific Corps project or program in place. It considers expected actions that may be executed by others, including potential future land use conditions, and shall consider effects of climate change using multiple scenario analyses.
 - (5) To ensure that the appropriate criteria and problems are incorporated into the analytical framework, a summary of the process used to define the relevant existing conditions and foreseeable future conditions shall be prepared and made available to the public and shared with stakeholders.
- (h) ***Formulate alternatives.*** The primary goal of an alternative is to address a water resources challenge, consistent with the Federal Objective and Guiding Principles. The primary function of an alternative must be to alleviate unsatisfactory conditions or address a problem or opportunity that exists or will exist in the future without the proposed Federal investment that is under consideration. Alternatives should address the defined water resources challenge that is the subject of the analysis, and may achieve multiple purposes. Alternative formulations should focus on solutions that are feasible and meet the planning objectives of the study, based on the most likely future conditions expected with and without implementation of an alternative. The viability of an alternative should be determined through an evaluation of its acceptability, efficiency, effectiveness, and completeness, as required in the PR&G. The period of analysis should be the same for each alternative and sufficient to encompass the lifespan and significant long-term impacts of the project. In addition, alternatives may also include actions which are beyond the missions of the Corps where others may help provide solutions to the identified problem and meet the goals of the PR&G. However, such alternatives shall identify the relevant parties with requisite responsibility for those actions beyond Corps missions (such as other Federal agencies and non-Federal

partners), their authority for that action, the interrelation between that action and the recommended Corps project, action, or program and appropriate sequencing of implementation. For Corps investments, the Corps will be the designated lead for completing PR&G analysis.

- (1) Alternatives are to be developed in a systematic manner that ensures that the Corps has identified and considered the full range of reasonable alternatives. A range of potential alternatives should be initially investigated reflecting a range of scales and measures, and as alternatives are refined, some would be screened out for reasons such as having excessive cost or unavoidable impacts, not sufficiently addressing the identified problem or opportunity, or other factors. The study report should include some analysis of the eliminated alternatives and reasons for their elimination. The plans that are retained for additional analysis will comprise the range of reasonable alternatives required for the NEPA analysis. Section 234.8 describes the alternatives required in the final array.
- (2) Consideration of nonstructural approaches and nature-based solutions that meet the planning objectives shall be an integral part of the development and evaluation of Federal investments in water resources.
- (3) Each alternative formulated for the PR&G analysis should align with the alternatives evaluated in the corresponding NEPA document.
- (4) The Corps should formulate the alternatives based on an incremental analysis of their benefits and costs to society. The economic, environmental, and social effects of a water resources development project are interrelated. In formulating alternatives to address the identified water resources problem or opportunity, the Corps shall consider each of these effects and seek to maximize net public benefits.

§ 234.7 Evaluation framework.

- (a) **General.** To inform the overall decision-making process, this section describes the common framework and general requirements to be used by the Corps in evaluating and ensuring full consideration of the social, environmental, and economic benefits and costs to society of any separable elements and potential alternatives for Federal investment. This will include their performance with respect to the Guiding Principles and their contributions to the Federal Objective. Any key assumptions that affect the analysis of alternatives shall be clearly described in the study.
- (b) **Economic, environmental, and social effects.**
 - (1) The Corps' analytical framework for evaluating Federal investments should focus on the key economic, environmental, and social effects that are relevant to the investment decision. Typical NEPA analyses emphasize environmental effects and benefits, including ecosystem services, and these should also be used as a core part of water resources alternatives analysis. A benefit-cost analysis would be conducted for each alternative. Ecosystem services are an important benefit-cost category that should be included in the benefit-cost analysis.
 - (2) In addition, the scale of the analysis can be adjusted for a given study. While all analyses should share common elements, how these elements are achieved can depend on the identified problem or opportunity. It is important to establish a consistent analytic approach for each study. When implementing its ASPs, the Corps will consider and, where it deems appropriate, align with the latest Federal methods and guidance (for example, updated OMB Circulars and applicable interagency guidance) to ensure that the analytical framework accounts for all significant economic, environmental, and social costs and benefits, including ecosystem services. Where possible, monetization enables the incorporation of the values placed on the benefits and costs evaluated and

provides a way to evaluate trade-offs in common analytical units (dollars). OMB Circulars A-4 and A-94 provide guidance on appropriate use of monetization methods. The Corps anticipates that it will not be possible to monetize all social and environmental costs and benefits of the alternatives. In these cases, the Corps should quantify the social and environmental costs and benefits and when neither monetization nor quantification is possible, the Corps should qualitatively describe the social and environmental costs and benefits in sufficient detail to allow differentiation across alternatives. Relevant monetary, quantitative, and descriptive information will be fully assessed and considered in the analysis.

(c) ***Best available actionable science and commensurate level of detail.***

- (1) Analysis to support the evaluation of alternatives shall use the best available actionable science, to include Indigenous Knowledge, data, analytical techniques, procedures, models, and tools in ecology, hydrology, economics, engineering, biology, and other disciplines to the extent that sufficient funding is available, and to the extent such information is relevant and appropriate to the subject investment. To the extent feasible, the effects of the alternatives should be monetized. Effects will be monetized, quantified, or described, in that order.
- (2) The level of detail required to support alternatives analysis may vary but should be sufficient to inform the decision-making process efficiently and effectively. The level of detail, scope, and complexity of analyses should be commensurate with the scale, impacts, costs, scientific complexities, uncertainties, risk, and other aspects (such as public concern) inherent in potential decisions.

(d) ***Risk and uncertainty.*** When analyzing potential Federal water resources investments, the Corps shall identify, describe, and quantify (if feasible) areas of risk and uncertainty and consider them in decision-making. Risks and uncertainties shall be identified and described in a manner that is clear and understandable to the public and decision-makers. This includes describing the nature, likelihood, and magnitude of risks, as well as the uncertainties associated with key supporting data, projections, and evaluations of competing alternatives. Risk and uncertainty are inherent in economic analyses as well as the analysis of physical and biological factors, no matter the technique or methodology employed. The study should estimate the extent to which the outcomes from an investment may vary over time from the estimates provided in the study, both in the short-term and in the less predictable future, due to uncertainty. Knowledge of risk and uncertainty and the degree of reliability of the estimated consequences will better inform decision-making. When there are considerable uncertainties concerning the ability of an alternative to function as desired (such as producing the desired outputs and/or the general acceptability of the alternative) the option of pursuing improved data or models should be considered. Reducing risk and uncertainty may involve increased costs or loss of benefits. The advantages and costs of reducing risk and uncertainty should be explicitly considered in formulating alternatives and the overall decision-making process.

(e) ***Adaptive management.*** Adaptive management measures shall be clearly identified and evaluated as part of alternatives to the extent that such measures are commensurate with the significance of the proposed activity and available resources. Adaptive management measures are particularly useful when making management choices in the face of uncertainty, such as when detailed information and tools are not readily available.

(f) ***Climate change.*** Conditions resulting from a changing climate shall be identified and accounted for in all stages of the planning process, and uncertainties associated with climate change will be identified and described. Analysis of climate change impacts shall reflect the best available actionable science and will leverage region-specific information from Federal, Tribal, State, local, and non-governmental partners. The

Corps shall incorporate a climate-informed science approach considering impacts such as inland and coastal climate change impacts on flood and drought hazards using the most up-to-date science, policies, and tools available. The Corps shall also ensure climate resilience and adaptation are incorporated and considered throughout the planning process and across alternatives, including a discussion on how climate, drought, and ecosystem resilience may intersect for that particular action and can contribute to the economic vitality and water resources resilience of the Nation. The changing climate should inform the understanding of water resources needs and how those needs can potentially be addressed.

- (g) **Water availability, water uses, and resilience.** The consideration of multiple uses, competing demands, and the potential future uses of the water resources shall be taken into account when formulating and evaluating solutions to a water resources problem or challenge. Water availability and efficient use of water shall be considered in designing the alternatives, as shall resilience, when applicable to the purpose of the study. The analysis shall consider water availability, water uses, and resilience over a range of conditions, from too little water in drought and multiple-use scenarios, to too much water in flood scenarios.
- (h) **Nonstructural and nature-based solutions.** Nonstructural measures alter the use of existing infrastructure or human activities to generally improve, avoid, or minimize adverse changes to existing hydrologic, geomorphic, and ecological processes. Nonstructural measures may be combined with fewer or smaller traditional structural project components to produce a complete alternative plan or may be used instead of a structural project. In the development of alternatives, the use of natural systems, ecosystem processes, and nature-based solutions shall be considered. Full consideration and reporting on nonstructural and nature-based alternative actions shall be an integral part of the evaluation of Federal water resources investment alternatives, and a full nonstructural in addition to a full nature-based alternative will be included in the final array of alternatives. Nonstructural and nature-based aspects should also be included in the other alternatives in the final array when appropriate.
- (i) **Tribal treaty rights.** Each of the alternatives considered for a water resources investment must be consistent with the protection of Tribal treaty rights. Alternatives that impact Tribal treaty and water rights should be screened out as soon as the Corps becomes aware of such impacts, and the study will document why the alternatives have been screened.
- (j) **State water law.** Alternatives for water resources investments must be consistent with applicable State laws, including water rights and decrees, to the extent that these do not conflict with Federal laws and regulations. Analyses should identify legal constraints that preclude selection of an otherwise viable alternative.
- (k) **International obligations.** Alternatives for water resources investments must be consistent with meeting treaty and other international obligations. Analyses should identify international obligations that preclude selection of an otherwise viable alternative.
- (l) **Period of analysis.** The period of analysis for the study shall be documented with an appropriate justification, and used to evaluate each alternative.

§ 234.8 Final array of alternatives.

- (a) Except as provided in paragraphs (c) and (f) of this section, the final array of alternatives shall include, at a minimum, the following six alternatives:
 - (1) A No Action alternative.

- (2) A nonstructural alternative: An alternative that can effectively address the problem through the feasible use of nonstructural approaches.
 - (3) A nature-based solution alternative: An alternative that can effectively address the problem through the feasible use of nature-based solutions (including natural systems and ecosystem processes).
 - (4) An environmentally preferred alternative.
 - (5) An alternative that seeks to maximize net public benefits to society. In developing this alternative, the Corps shall not consider regional economic development effects.
 - (6) An alternative that is locally preferred. If this alternative differs from the net public benefits alternative, it will be required to have a comparable level of detail and analyzed using the same analytical framework as the net public benefits alternative.
- (b) The nonstructural and nature-based alternatives do not preclude consideration of these elements in other alternatives. Nonstructural measures and nature-based solutions shall be considered as components of the other alternatives in the final array, essentially providing an integrated or “hybrid” of gray (hard) infrastructure with these other measures.
- (c) The same alternative may be identified as more than one of these required alternatives.
- (d) Mitigation of unavoidable adverse effects associated with each alternative must be included in the alternative and in the analyses.
- (e) If an alternative requires changes in existing laws, regulations, or policies, those changes must be clearly identified and explained.
- (f) If one or more of the required alternatives is not included because the Corps was not able to identify a potential solution that is feasible and consistent with the purpose of the study, the study must document that decision.
- (g) The discussion of the final array of alternatives should include the primary purpose of the analysis; the geographic size of the study area; the types of impacts; the number of people potentially affected and anticipated degree of impact; environmental justice considerations; the size and location of communities potentially affected including the presence of Federally-recognized Tribes or Tribal members; and the type of data and information available from Indigenous Knowledge, collaboration, public involvement, and previous studies.

§ 234.9 Evaluate effects of alternatives.

- (a) **Analysis of alternatives.** For the final array of alternatives, the analysis should describe, evaluate, and estimate the key social, environmental, and economic effects, and assess the contributions of each alternative to the Guiding Principles. The analysis should identify any impacts to Tribal treaty rights which were unknown earlier in the planning process and which prevent the selection of an alternative.
- (b) **Evaluation procedures.** In addition to assessing how alternatives perform with respect to the Guiding Principles, the evaluation procedures shall incorporate methods to evaluate:
- (1) How public benefits of an alternative compare to its costs, including full consideration of all important social, environmental, and economic benefits and costs.
 - (2) How alternatives perform against the objectives of the study.

- (3) How alternatives perform against the four formulation criteria: completeness, effectiveness, efficiency, and acceptability.

(c) **Consideration of benefits and costs.** The report should fully account for the effects to society of alternative plans and their respective contributions to the Federal Objective, relative to the No Action alternative. The analysis will evaluate the economic benefits and costs, environmental benefits and costs, and social benefits and costs of alternatives, regardless of how they are included (monetized, quantified, or described). The consideration of benefits to federally recognized Tribes will be done in direct consultation with the respective Tribal Nation and will supplement the public benefit analysis. To the extent practicable, such costs and benefits must be quantified in a scientifically valid and acceptable way, and such quantified costs shall be monetized where practicable. When monetization or quantification is not possible, costs and benefits must be described in sufficient detail to enable the decision-maker to understand the importance and magnitude of potential changes. For monetized costs and benefits, the present value cost of each alternative must be compared to the present value of the benefit to the public. For quantified but not monetized benefits and costs, the Corps would present the information on an average annual basis, and would also describe how the benefits and costs would accrue over the period of analysis. For qualitatively described benefits and costs, expectations would be described across the period of analysis. The effects of alternative plans are displayed in terms of costs and benefits.

§ 234.10 Compare alternatives.

(a) **Comparing alternatives.** Alternatives shall be compared to each other and to the No Action alternative and shall include a comparison of the ability of the alternatives to perform under changing conditions, including climate change. The alternative (or alternatives) that reasonably meets the Federal Objective and maximizes net public benefits shall be identified. In addition, alternatives may be evaluated separately with respect to other considerations, including distributional effects. These considerations may include:

- (1) Temporal factors, since certain effects may occur at different points in time.
- (2) Spatial factors, since certain costs, benefits, and transfers may accrue to different regions. Regional-scale analyses may be useful to inform regional-level economic development objectives. It is important to note that such regional analyses, while useful, are completely separate from the calculation of net public benefits described in § 234.4(c).
- (3) Beneficiaries. Tribal Nations and stakeholders (including other governmental agencies and communities with environmental justice concerns) may indicate different tradeoffs among the various benefits and costs of a Federal action. Tribal reserved rights, including treaty-protected resources and habitats, are not benefits to the Tribal Nation, rather, they are guaranteed by such treaties. Robust engagement at this stage shall focus on eliciting preferences among the alternatives, their component elements, and their effects. When calculating net benefits, these distributional effects can be examined using techniques like income weighting.

(b) **Tradeoffs.** Tradeoffs among potential alternatives will be assessed and described throughout the decision-making process and in a manner that informs decision-making. Based on the available analytical information, the Corps would use its professional judgment in making its recommendations on decisions among tradeoffs. The tradeoff displays shall be understandable, transparent, and constructed in a generally consistent fashion for all analyses. The analysis shall include a combination of both tables and explanatory materials to help inform a decision. Displays shall facilitate the evaluation and comparison of alternatives necessary to make the following determination and reflect the following:

- (1) The effectiveness of alternatives in solving the water resources problem and taking advantage of the opportunities identified in the planning process.
 - (2) What must be given up in monetary and nonmonetary terms to enjoy the benefits of the various alternatives, relative to the baseline.
 - (3) The differences among alternatives.
- (c) **Information for inclusion in the analysis.** To promote consistency across the Corps, the following tables and information shall be included in the analysis and documentation prepared for a decision process:
- (1) **Criteria.** The analysis must explicitly address the extent to which an alternative achieves each of the following criteria: completeness, effectiveness, efficiency, and acceptability. This evaluation must be systematic and can include both quantitative and qualitative components.
 - (2) **Effects matrix.** A matrix summarizing the tradeoffs, relative to the baseline, effect-by-effect must be included in the integrated report.
 - (3) **Additional trade-off displays.** Additional text and tables should display other important trade-offs, such as trade-offs along temporal, spatial, and beneficiary dimensions.
 - (4) **Summary table.** A summary table displaying the economic, environmental, and social costs and benefits as measured (monetized, quantified, quantitative) for each alternative. In addition, the summary table will display the economic, environmental, and social costs and benefits which were derived qualitatively. The summary table will also separately include information on the level of risk or uncertainty for each alternative.
 - (5) **Achievement of objectives table.** A table indicating the extent to which the Guiding Principles have been achieved.

§ 234.11 Select the recommended plan.

- (a) **Recommended plan.**
- (1) Plan selection will require decision-makers to assess tradeoffs and to consider the extent of both monetized and non-monetized effects. The basis for selection of the recommended plan should be fully reported and documented in a transparent manner, including the criteria and considerations used. This section must provide a discussion about the extent to which the alternatives achieve the Federal Objective and maximize net public benefits to society. The report must include an explanation of the assumptions in the evaluation of monetized and non-monetized benefits and costs. This section will include a summary of elicited Tribal Nation and stakeholder perspectives on the alternatives and their effects.
 - (2) The Corps should recommend:
 - (i) Authorization of an alternative project, program, or plan;
 - (ii) Implementation of an alternative under existing law;
 - (iii) Implementation of a project, program or plan by others; or
 - (iv) No action.
 - (3) In its studies, the Corps shall analyze, evaluate, fully consider, and justify each separable element of the proposed investment independently of the other separable elements, based on its social, environmental, and economic benefits and costs to society.

(4) The Corps should seek to meet water resources objectives and maximize net public benefits, relative to public costs. It is possible that more than one alternative might reasonably and approximately meet these conditions. "Net public benefits" implies that the anticipated benefits will be presented relative to the costs associated with the accrual of those benefits. Net public benefits can include both quantified and non-quantified benefits. Any recommendation will clearly delineate the Federal water resources project(s) or actions being recommended, including any condition precedent for construction.

(b) **Exceptions.** A recommended plan for a Federal water resources investment that does not maximize net public benefits requires an exception from the Assistant Secretary of the Army for Civil Works. Requests for exception should describe the project or activity, the rationale for the exception, and present relevant data and analysis to support the request.